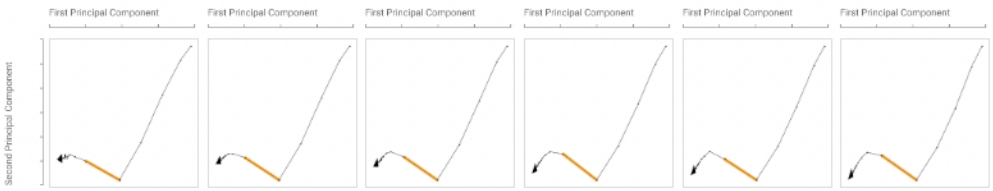
FULL-SCALE TRANSFORMERS

This experiment series analyzes a sweep of typical transformers, spaced exponentially in size, up to a 13 billion parameter model. The goal is to show that the phenomena we observed in small models can be seen in larger models as well. For these models, we unfortunately have lower time resolution (note that the x-axis of most plots has changed to a log scale). We are also unable to provide complete ablations. The main observation to make is that results are roughly the same as seen in the small models. More...

4-LAYER TRANSFORMER (13M PARAMS) | 6-LAYER TRANSFORMER (42M PARAMS) | 10-LAYER TRANSFORMER (200M PARAMS) | 16-LAYER TRANSFORMER (810M PARAMS) | 24-LAYER TRANSFORMER (2.7B PARAMS) | 40-LAYER TRANSFORMER (13B PARAMS)

PCA OF TOKEN LOSSES

The vector of per-token losses is a way to map different neural network behavior to the same vector space. We take 10,000 individual token predictions per model, and project them onto the first two principal components. This shows how the large-scale-behavior of multiple networks evolve over training. More...

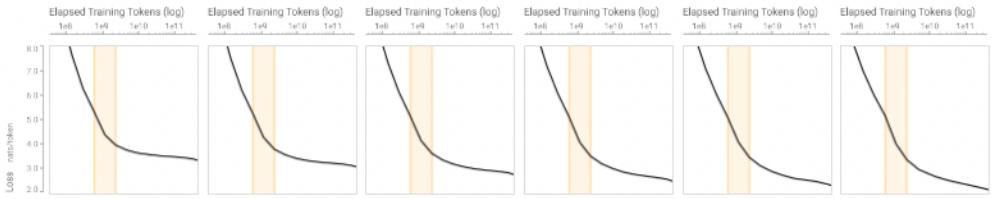


MODELS OVER TRAINING

LOSS OVER TRAINING

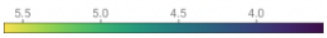
Loss curve showing $\mathrm{Loss}(n_{\mathrm{train}})$ after the model is trained on n_{train} tokens of training data. More...

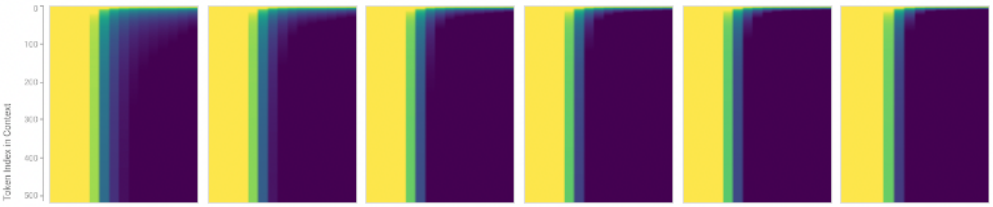
The orange band denotes phase change interval.



LOSS OVER TRAINING, BROKEN DOWN BY CONTEXT INDEX

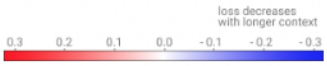
Heatmap of $\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})$, the average loss of i_{ctx} token in context after n_{train} elapsed tokens of training. More...

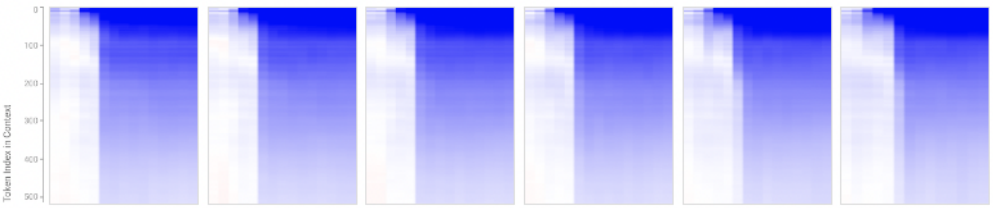




DERIVATIVE OF LOSS WRT CONTEXT INDEX

Heatmap of $d\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})\ /\ d\ln(i_{\mathrm{ctx}})$. The log vertical partial derivative of the above graph. More...

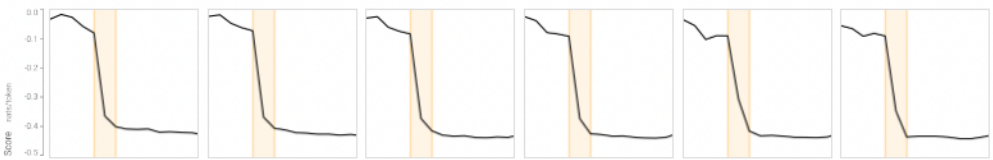




"IN-CONTEXT LEARNING SCORE"

$\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 500)$
$- \mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 50))$

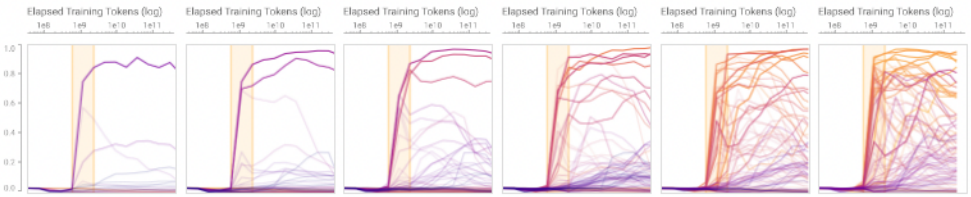
How much better the model is at predicting the 500th token than the 50th? We use this as a proxy for how good the model is at in-context learning. More...



ATTENTION HEADS OVER TRAINING

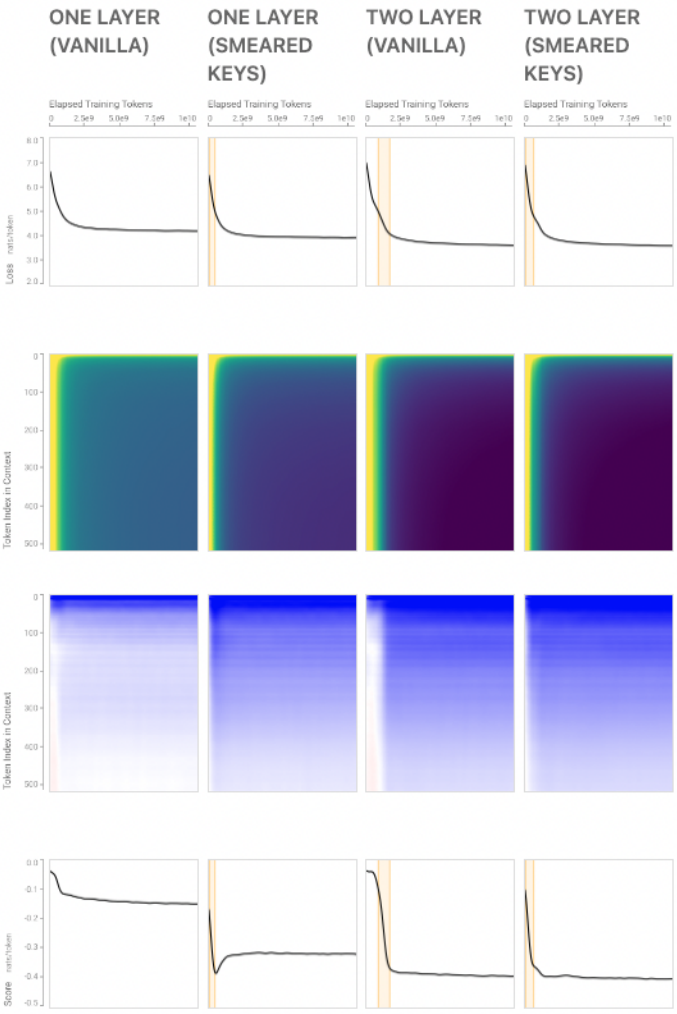
"PREFIX MATCHING SCORE"

Each line is an attention head. Y axis is the average fraction of a head's attention weight given to the token we expect an induction head to attend to -- the token where the prefix matches the present context -- on random, synthetic data. Heads are colored by layer depth: early; late. More...



"SMEARED KEY" ARCHITECTURE MODIFICATION + CONTROLS

This experiment series analyzes "smeared key" models, which are designed so that a single attention layer can implement induction heads (see Argument 2). For each smeared key model, a regular control model is provided (trained exactly the same) for comparison. The key observations are that (1) a one-layer smeared-key model experiences the phase change, while a regular one-layer model does not; and (2) the two-layer smeared-key model experiences a phase change earlier than a baseline model. More...



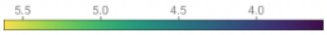
LOSS OVER TRAINING

Loss curve showing $\mathrm{Loss}(n_{\mathrm{train}})$ after the model is trained on n_{train} tokens of training data. More...

The orange band denotes phase change interval.

LOSS OVER TRAINING, BROKEN DOWN BY CONTEXT INDEX

Heatmap of $\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})$, the average loss of i_{ctx} token in context after n_{train} elapsed tokens of training. More...



DERIVATIVE OF LOSS WRT CONTEXT INDEX

Heatmap of $d\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})\ /\ d\ln(i_{\mathrm{ctx}})$. The log vertical partial derivative of the above graph. More...

"IN-CONTEXT LEARNING SCORE"

$$\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 500) - \mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 50))$$

How much better the model is at predicting the 500th token than the 50th? We use this as a proxy for how good the model is at in-context learning. More...